

# Jooho Kim

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## Education

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| <b>University of Minnesota, Twin Cities</b><br><i>PhD in Statistics</i>                                                                 | Minnesota, United States<br>Sep. 2026 –       |
| <b>Seoul National University</b><br><i>MS in Statistics</i>                                                                             | Seoul, South Korea<br>Mar. 2024 – Feb. 2026   |
| <b>Korea University</b><br><i>BE in Food and Resource Economics, Double major in Statistics</i><br><i>Military Service, 2020 – 2021</i> | Seoul, South Korea<br>Mar. 2018 – Feb. 2024   |
| <b>The University of Texas at Austin</b><br><i>Exchange Program, Economics</i>                                                          | Texas, United States<br>Aug. 2022 – Dec. 2022 |

## Research Interests

Missing Data, Data Integration, Statistical Machine Learning, High-Dimensional Inference

## Preprints

**Kim, J.**, Saegusa, T., and Shin, Y. E. (2026+). “Scalable and Efficient Multiple Imputation for Influence-Based Supersampled Case-Cohort Studies.” Revision invited at *Biometrics*. [\[preprint\]](#) [\[code\]](#)  
*Best Student Oral Presentation Award, Korean Statistical Society.*

## Research Experience

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| <b>Prediction Model Lab, Seoul National University</b><br><i>Graduate Researcher (Advisor: Dr. Yeí Eun Shin)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Seoul, South Korea<br>Jun. 2024 – Present      |
| <ul style="list-style-type: none"> <li>◦ Led as the primary graduate researcher on a project funded by the National Research Foundation of Korea: “<i>Multiple Imputation for Missing Covariates due to Epidemiological Sampling Designs</i>”.</li> <li>◦ Proposed an influence function-based supersampling approach that imputes only a subset (e.g., 3%) of the missing covariates while preserving efficiency and unbiasedness.</li> <li>◦ Devised weight calibration equations that integrate distinct sampling designs for a unified analysis.</li> <li>◦ Applied the proposed method to the NIH–AARP Diet and Health Study to assess expensive biomarkers associated with pancreatic cancer risk using the Cox proportional hazards model.</li> </ul> |                                                |
| <b>Urban Informatics Lab, The University of Texas at Austin</b><br><i>Undergraduate Research Assistant (Connected through Dr. Arya Farahi)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Austin, United States<br>Oct. 2022 – Dec. 2022 |
| <ul style="list-style-type: none"> <li>◦ Aggregated and cleaned geotagged electric vehicle (EV)-related tweets across the U.S. using regular expressions and bot probability scores.</li> <li>◦ Conducted hotspot analysis to identify regions with significant EV-related public sentiment.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                |

## Contributed Talks

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| <b>Scalable and Efficient Multiple Imputation for Case-Cohort Studies via Influence Function-Based Supersampling</b><br>Korean Statistical Society, Seoul, Korea.                | Dec. 2025 |
| <b>Multiple Imputation for Incomplete Survival Data with Missing Covariates: Toward Valid Causal Inference</b><br>The 2nd Symposium on Causal Inference, Seoul, Korea (English). | Jun. 2025 |

## Honors and Awards

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| <b>Best Student Oral Presentation Award</b><br><i>Awarded by the Korean Statistical Society (KSS) for a graduate student paper presentation.</i>                       | Dec. 2025                          |
| <b>Fellowship for Fundamental Academic Fields</b><br><i>Awarded by Seoul National University for academic excellence and research potential.</i>                       | May 2024; Feb. 2025                |
| <b>Graduate Research Fellowship in Science and Engineering</b><br><i>Awarded by the National Research Foundation of Korea through a competitive selection process.</i> | Sep. 2024 – Aug. 2025              |
| <b>Special Scholarship</b><br><i>Awarded by Korea University for academic excellence.</i>                                                                              | Sep. 2022; Mar. 2023               |
| <b>Semester High Honors</b><br><i>Recognized by Korea University for academic excellence.</i>                                                                          | Sep. 2018; Mar. 2022;<br>Mar. 2023 |
| <b>Agricultural Economics Alumni Scholarship</b><br><i>Awarded by Korea University, Department of Food and Resource Economics for academic excellence.</i>             | Mar. 2022                          |

## Teaching Assistantship

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| <b>Survival Data Analysis and Lab</b><br><i>Advanced Undergraduate Course</i> <ul style="list-style-type: none"><li>◦ Led hands-on lab sessions on survival analysis and graded assignments and exams.</li></ul> | Fall 2025   |
| <b>Selected Topics Seminar</b><br><i>Introductory Undergraduate Course</i> <ul style="list-style-type: none"><li>◦ Organized weekly discussion sessions and advised on data analysis projects.</li></ul>         | Spring 2025 |
| <b>Mathematical Statistics 2</b><br><i>Core Undergraduate Course</i> <ul style="list-style-type: none"><li>◦ Held office hours, graded assignments and exams, and prepared solution sets.</li></ul>              | Fall 2024   |
| <b>Statistics Lab</b><br><i>Introductory Undergraduate Course</i> <ul style="list-style-type: none"><li>◦ Evaluated Python programming assignments and exams and held office hours.</li></ul>                    | Spring 2024 |

## Employment

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| <b>Hankuk University of Foreign Studies – Insight Camp</b><br><i>Academic Mentor</i> <ul style="list-style-type: none"><li>◦ Provided residential academic mentoring and taught math and English classes in a 4-week program.</li><li>◦ Created problem-solving exercises and offered individualized tutoring.</li></ul> | Jan. 2019 – Feb. 2019 |
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## Projects

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| <b>Survey Design and Analysis for the SNU Panel of Youth (SNUPY)</b> <ul style="list-style-type: none"><li>◦ Constructed stratified cross-sectional and longitudinal sampling weights using the R <code>survey</code> package.</li><li>◦ Advised on missing data imputation and weight calibration.</li></ul> | Oct. 2024; Dec. 2025 |
| <b>Modeling Risk Factors for Mortality and Hospitalization</b> <ul style="list-style-type: none"><li>◦ Analyzed clinical risk factors for mortality and hospital stay using GLMM and multiple imputation, addressing repeated events and missing data.</li></ul>                                              | Sep. 2025            |
| <b>Bitcoin Chart Pattern Image Recognition and Price Prediction</b><br><a href="#">GitHub Repository</a> <ul style="list-style-type: none"><li>◦ Implemented Monte Carlo Dropout in the N-BEATS time-series neural network to quantify and visualize predictive uncertainty.</li></ul>                        | May 2022 – Jul. 2022 |

- Augmented chart image data using probability distributions, resulting in a 10% increase in accuracy.

**Optimizing Pricing Strategies for a Low-Demand Food Product**

May 2022 – Jun. 2022

- Designed an online survey and conducted a conjoint analysis to identify consumer preferences.
- Developed a Python algorithm to estimate the profit-maximizing bundle price for the food product.

**Data Visualization of Job Openings in Korea**

Nov. 2021 – Jan. 2022

[GitHub Repository](#) (In Korean)

- Extracted and preprocessed 36,000 job postings and 11,000 resumes by identifying HTML patterns.

## **Skills & Languages**

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**Software** R, Python, LaTeX, SAS, ArcGIS, Stata, SPSS

**Languages** Fluent in both English and Korean